TRIPBUDDY PROJECT

Table of Contents:

1. Team Introduction: **3**

2. Case Study: TripBuddy: **3**

3. Business Analysis / System Design: **4**

3.1. Business Analysis: 4

3.2. Stakeholders: 4

3.3. Functional Requirements: 4

3.3.1. User Authentication: 4

3.3.2. Recommendation Features: 5

3.3.3. Trip Planning: 5

3.3.4. Notifications: 5

3.3.5. User Profile: 5

3.4. System Design & Architecture: 5

3.4.1. Architecture Overview: 5

3.4.2. System Components: 6

3.4.3. Functional Flow Description: 6

a) Login/Signup Flow: 6

b) Recommendation Flow: 7

c) Review Flow: 7

d) Itinerary Flow: 7

e) Notification Flow: 7

3.4.5. Database Design: 7

a) User Collection: 7

b) Location Collection: 8

c) Evaluation Collection: 8

d) Schedule Collection: 8

e) Notification Collection: 9

3.4.6. Technologies Used: 9

4. Demo of your Android application: **9**

4.1. Welcome Screen (Splash Screen): 9

4.2. Login Screen: 9
4.3. Signup Screen: 10
4.4. Recommended Attractions - List: 10
4.5. Recommended Attractions - Map View: 11
4.6. Attraction Details Screen: 11
4.7. Review Location - List: 11
4.8. Review Location - Submit Review: 12
4.9. Schedule - Overview: 12
4.10. Schedule - Add Event: 13
4.11. Notification List: 13
4.12. Notification Details: 13
4.13. User Profile & Settings: 14
4.14. Travel History: 14
4.15. About Us: 15
4.16. QnA: 15
5. Conclusion and Discussion: 16
5.1. Conclusion: 16
5.2. Pros of the Application: 16
5.2. Cons of the Application: 17
5.3. Lessons Learned: 17
5.4. Future Improvements: 18
6. Contribution: 18
7. Handover Submission: 19

1. Team Introduction:

We are a group of five developers working on creating solutions that enhance the travel experience. Our team consists of:

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Together, we aim to build innovative mobile applications that connect people and bring value to everyday life.

2. Case Study: TripBuddy:

TripBuddy is a mobile application designed to connect travelers with local residents to discover hidden and lesser - known tourist spots. The platform addresses the growing demand for unique and affordable travel experiences that mainstream apps often overlook. While most travel platforms suggest popular tourist destinations, TripBuddy allows local people to share affordable, lesser - known places that only they are familiar with. Travelers can search for recommended spots, read reviews, plan their itineraries, and get personalized suggestions based on their location. Locals can contribute by sharing their favorite spots, and users can review and rate locations, helping others discover the best places to visit.

- Business Rules:
    - User Registration: Both travelers and locals must sign up to access app features.

  - Recommendation System: Local residents can post recommendations for specific places (attractions, restaurants, etc…).
  - Reviews: Users can leave ratings (1-5 stars) and write reviews for visited locations.
  - Schedule Planning: Travelers can create and manage itineraries with recommendations.

- Notifications: Users receive alerts for new recommendations, reviews, or itinerary updates.

3. Business Analysis / System Design:

3.1. Business Analysis:

Problem Statement: The travel industry is saturated with apps suggesting popular tourist spots, but these often lack information about hidden, budget-friendly, and culturally rich destinations that locals know best. TripBuddy aims to bridge this gap by connecting travelers with local experts, allowing travelers to discover these hidden gems and plan their trips with personalized recommendations.

3.2. Stakeholders:

- Travelers: Users looking for unique and affordable travel destinations.
- Local residents: Users contributing local knowledge by sharing recommendations and reviews.
- Platform Admin: Manage users, content, and ensure compliance with platform guidelines.

3.3. Functional Requirements:

3.3.1. User Authentication:

- Login/Signup: Secure login/signup using email and password.

- Password Management: Forgot password, password reset.

3.3.2. Recommendation Features:

- View Recommendations (List & Map): Travelers can view a list of recommended destinations and see them on a map.
- Filters and Sorting: Filter recommendations based on location, rating, category (e.g., restaurants, sightseeing), and budget.
- Post Recommendation: Locals can post new recommendations, including uploading images, descriptions, and tagging locations on the map.
- Review and Rate Recommendations: Travelers can review and rate destinations after visiting them.
- Attraction Details: View detailed information on attractions, including images, descriptions, ratings, and reviews.

3.3.3. Trip Planning:

- Create Itinerary: Travelers can add locations to their trip schedule and organize them by day and time.
- Edit Itinerary: Modify existing schedules and trips as needed.
- Reminders: Notifications for upcoming scheduled trips or events.

3.3.4. Notifications:

- Real - time Notifications: Travelers receive updates when a new review is posted on places they have visited or when a local submits a new nearby recommendation.

3.3.5. User Profile:

- View/Edit Profile: Users can update their personal information, view history of visited/reviewed locations, and manage notification settings.

3.4. System Design & Architecture:

3.4.1. Architecture Overview:

- Client-Server Model with REST API.

- Mobile App: Android, developed in Java (UI in XML).
- Backend: Built with C# using ASP.NET Core.
- Database: SQL Server for structured data storage.
- Cloud: Hosted on Azure Cloud.
- Map Integration: Uses Google Maps API.

3.4.2. System Components:

- Mobile Client (Android):
  - Developed using Java with XML for UI.
  - Communicates with backend via REST API.
  - Key features: login/signup, recommendations, reviews, and itineraries.
- Backend (C# with ASP.NET Core):
  - Handles business logic: user management, recommendations, reviews, and itineraries.
  - Secure API communication with JWT authentication.
  - Hosted on Azure Cloud.
- Database (SQL Server):
  - Stores data.

3.4.3. Functional Flow Description:

a) Login/Signup Flow:

- User accesses the Login Screen.
- The system checks if the user has an account. If yes, proceed to login. If no, direct to the Signup Screen.

- After authentication, the user is redirected to the Recommended Attractions - List screen.

b) Recommendation Flow:

- Travelers or locals can post new recommendations via the Post Recommendation screen, uploading images and descriptions.
- These recommendations are stored in the Locations collection and are made available to other users in the Recommended Attractions - List and Map View.

c) Review Flow:

- Travelers can visit the Review Location screen after visiting a place.
- They can submit a rating and review, which is stored in the Reviews collection.
- Reviews are visible to other users when they view the Attraction Details of that specific location.

d) Itinerary Flow:

- Users can add recommendations to their personal trip planner through the Schedule - Add Event screen.
- This data is saved in the Schedules collection, and reminders can be set to notify the user before the event.

e) Notification Flow:

- Notifications are generated when there are updates, such as new reviews or suggestions in the vicinity.
- These notifications are sent via Firebase Cloud Messaging to the user's device.

3.4.5. Database Design:

a) User Collection:

- userId: Unique identifier for each user.

- name: Full name of the user.
- email: Email for login.
- passwordHash: Securely stored hashed password.
- role: Traveler or Local.
- profilePicture: URL to profile image.
- notificationPreferences: Settings for push notifications.

b) Location Collection:

- locationId: Unique ID for the location.
- name: Name of the place.
- description: Description of the location.
- imageUrls: List of image URLs.
- postedBy: User who posted the recommendation.
- ratings: Average rating based on user reviews.
- coordinates: Latitude and longitude for map display.

c) Evaluation Collection:

- evaluationId: Unique ID for the review.
- userId: ID of the reviewer.
- locationId: ID of the location being reviewed.
- rating: Star rating (1-5).
- comment: Written review.
- imageUrls: Optional images uploaded with the review.

d) Schedule Collection:

- scheduleId: Unique ID for the schedule entry.
- userId: ID of the user who created the schedule.
- events: List of events (places and times) scheduled by the user.
- reminderTime: Optional field for reminder notifications.

e) Notification Collection:

- notificationId: Unique notification ID.
- userId: User to whom the notification is sent.
- message: The content of the notification.
- isRead: Boolean to track if the notification is read.

3.4.6. Technologies Used:

- Back-end: ASP .NET API.
- Front-end: XML, Code Behind: Java.
- Database: SQL Server.
- Hosting: Azure Cloud.

4. Demo of your Android application:

4.1. Welcome Screen (Splash Screen):

- Function: Displays the app logo and slogan for a few seconds when the app is opened, creating a welcoming first impression.
- Explanation: This screen serves as a brief introduction to the app, allowing the app's resources to load in the background before redirecting the user to the login screen.
    + Action: After a few seconds, it transitions to the Login Screen.

4.2. Login Screen:

- Function: Users enter their email and password to log in to the app.

- Explanation: Authentication ensures that users can securely access their accounts and personalized data, such as saved itineraries or previous reviews.
    + Features:
        - Email & Password fields.
        - Login Button: Redirects to the Recommended Attractions - List if credentials are correct.
        - Forgot Password Link: Allows users to reset their password.
        - Signup Link: Directs new users to the Signup Screen.

4.3. Signup Screen:

- Function: Allows new users to create an account by entering their email, password, and name.
- Explanation: New users need an account to fully access the app's features, including posting recommendations and saving itineraries.
    + Fields:
        - Email, Password, and Name fields.
        - Sign up Button: Creates the user account and redirects to the Recommended Attractions - List.

4.4. Recommended Attractions - List:

- Function: Displays a list of recommended attractions shared by local residents.
- Explanation: This is the core feature of the app where users can explore hidden gems suggested by locals. Users can filter recommendations by location, type (e.g., restaurant, sightseeing), and rating.
    + Features:
        - A search bar and filters to refine suggestions based on location, budget, and type.
        - Tapping on a recommendation leads to the Attraction Details Screen.

4.5. Recommended Attractions - Map View:

- Function: Displays a map showing the recommended attractions pinned by their locations.
- Explanation: Users can visualize the exact locations of attractions on a map, making it easier to navigate and explore nearby recommendations.
    + Features:
        - Google Maps integration: Allows users to zoom in/out and tap on pins to view details about each attraction.
        - Tapping a pin brings up a preview card with the name of the attraction and a link to the Attraction Details.

4.6. Attraction Details Screen:

- Function: Displays detailed information about a selected attraction.
- Explanation: Provides users with in-depth descriptions, reviews, and images of the attraction, allowing them to make informed decisions on whether to visit.
    + Details:
        - Images: A carousel of images uploaded by the user or other visitors.
        - Description: Text describing the place.
        - Reviews: A list of reviews with user comments and star ratings.
        - Post Review Button: Links to the Review Location - Submit Review Screen.

4.7. Review Location - List:

- Function: Displays a list of all reviews for a specific location.
- Explanation: Users can view comments and ratings left by other visitors, which helps gauge the quality and experience of the place.

+ Features:
    - Review Cards: Each card contains the user's rating, written review, and any images they attached.
    - Submit Review Button: Redirects users to the Submit Review Screen if they want to contribute their own review.

4.8. Review Location - Submit Review:

- Function: Allows users to submit a review for a location they've visited.
- Explanation: Gathering user-generated content ensures the recommendations remain trustworthy and up-to-date.
    + Fields:
        - Star Rating (1-5): Users can rate their experience.
        - Review Textbox: Users can write their thoughts and feedback.
        - Attach Image: Option to upload pictures of the location.
        - Submit Button: Posts the review, which becomes visible to other users.

4.9. Schedule - Overview:

- Function: Displays the user's planned itineraries.
- Explanation: This feature helps travelers organize their trips by adding scheduled visits to recommended places, events, or personal reminders.
    + Features:
        - Day/Week/Month View: Allows users to view their itinerary based on time intervals.
        - Each entry shows the location name, date, and time.
        - Tap on an event to see details or make edits.
        - Add New Event Button: Links to the Schedule - Add Event Screen.

4.10. Schedule - Add Event:

- Function: Allows users to add new events or destinations to their itinerary.
- Explanation: Users can schedule visits to specific locations or add custom events, ensuring they can plan their trip efficiently.
    + Fields:
        - Event Name: Name of the place or event.
        - Location: Select a recommended location or manually enter one.
        - Date and Time: Set when the event will occur.
        - Reminder: Option to receive notifications before the event.
        - Save Button: Adds the event to the user's itinerary.

4.11. Notification List:

- Function: Displays all notifications relevant to the user.
- Explanation: Keeps users updated on important actions like new reviews on places they've visited, local recommendations near them, or schedule reminders.
    + Features:
        - Each notification shows a summary (e.g., "New review on XYZ Restaurant").
        - Tap on a notification to see more details in the Notification Details Screen.

4.12. Notification Details:

- Function: Shows the full details of a selected notification.
- Explanation: Users can read more information about an alert, such as a full review or a trip reminder.
    + Features:
        - Displays the content of the notification in detail.

- Includes options to dismiss or take action (e.g., view location details, update itinerary).

4.13. User Profile & Settings:

- Function: Allows users to view and update their profile, manage notification settings, and access privacy options.
- Explanation: Gives users control over their personal information and app preferences.
    + Features:
        - Profile Information: Users can update their name, email, or profile picture.
        - Change Password: Users can update their password.
        - Notification Preferences: Toggle notifications for reviews, reminders, or new recommendations.
        - Privacy Settings: Options for handling public or private recommendations.
        - Logout Button: Logs the user out and redirects to the Login Screen.

4.14. Travel History:

- Function: Displays a list of the user's past trips.
- Explanation: Users can view details about previous travels, helping them recall experiences and plan future trips.
    + Fields:
        - Trip List:
            - Data Type: List (ListView or RecyclerView).
            - Content:

- Destination: Name of the destination.
- Start Date: Date of departure.
- End Date: Date of return.
- Trip Status: Completed, Ongoing, Canceled.
- Detail Button: Button to view trip details.

4.15. About Us:

- Function: Provides information about the app, development team, and contact details.
- Explanation: Users can learn about the app's purpose, the team behind it, and how to reach out for support.
    + Fields:

        - App Information.
        - Team Information.
        - Contact Information.
        - Social Media Links.
        - Terms and Conditions.

4.16. QnA:

- Function: Displays frequently asked questions and answers.
- Explanation: Helps users find quick answers to common queries about the app.
    + Fields:

        - Search Bar.
        - Question List.
        - Answer (displayed when the user expands a question).

- Contact Support.

5. Conclusion and Discussion:

5.1. Conclusion:

The TripBuddy application successfully bridges the gap between travelers and local experts, providing a platform where lesser-known and affordable attractions can be shared and discovered. It addresses a significant issue in the travel industry, where mainstream apps often highlight only popular tourist spots, while TripBuddy empowers locals to contribute insights about hidden gems. By offering features like personalized recommendations, itinerary management, and user reviews, TripBuddy delivers a user-centric solution that enhances the travel experience.

5.2. Pros of the Application:

- Unique Value Proposition: Unlike traditional travel apps, TripBuddy focuses on providing travelers with local, hidden, and budget-friendly recommendations, offering a more authentic experience.
- User-Generated Content: The app thrives on recommendations and reviews from locals, making the information fresh, relevant, and specific to each area.
- Interactive and User-Friendly Design: Features like viewing recommendations on a map, submitting reviews with images, and managing itineraries provide a seamless user experience.
- Itinerary Management: The ability to plan trips and receive reminders makes the app a useful tool for trip organization, further enhancing its practicality for travelers.
- Push Notifications: Real-time notifications keep users engaged by alerting them about new reviews, recommendations, or upcoming events.

5.2. Cons of the Application:

- Limited Initial Content: At the start, the app relies heavily on users (locals) to submit recommendations. Without a substantial initial user base, the app might struggle with content creation.
- Dependence on Internet Connectivity: The app relies on constant internet access for displaying recommendations, updating reviews, and interacting with the map. It may not be as useful in areas with limited internet connectivity.
- Scalability Issues: While the backend and database design (MongoDB, Firebase) are scalable, a sudden influx of users could strain the server infrastructure, leading to performance bottlenecks.
- No Offline Access: Currently, there's no offline mode for travelers who want to access their itineraries or saved locations without an internet connection.
- Lack of Advanced Personalization: The app provides recommendations based on location, but it lacks advanced algorithms to suggest places based on user preferences or travel history.

5.3. Lessons Learned:

- User Experience and Design: The importance of intuitive navigation and a clean interface became very clear during development. Balancing functionality with simplicity is key to making an app that users enjoy using.
- Backend Scalability: We learned how crucial it is to design a backend that can scale with the number of users and how cloud-based solutions like Firebase help manage authentication, notifications, and media storage efficiently.
- API Integration: Integrating with external services, such as Google Maps API for location-based features, was an insightful experience. This gave us a deeper understanding of how to utilize third-party APIs to add value to the app.

- Collaborative Development: Working as a team on a single project taught us the importance of version control, clear communication, and breaking down tasks efficiently to ensure smooth progress.

5.4. Future Improvements:

- Offline Mode: Adding an offline mode would be a priority to allow users to access saved itineraries and recommendations without an internet connection.
- Enhanced Recommendation Algorithm: Implementing a recommendation engine that personalizes suggestions based on user preferences, travel history, and reviews would significantly improve the user experience.
- Content Moderation and Verification: As the app grows, ensuring the accuracy and reliability of user-submitted content will be important. Introducing moderation features or a trust system (based on user reputation) would help maintain content quality.
- Social Features: Adding social features, such as following locals or fellow travelers, sharing itineraries, or creating travel groups, would increase user engagement and community interaction.
- Gamification: Introducing rewards or badges for frequent users (both travelers and locals) could incentivize content creation and enhance user engagement.
- Cross-Platform Expansion: While the current app is designed for Android, expanding to iOS would broaden the user base and make the app more accessible to a wider audience.

6. Contribution:

Topic	Team Effort	Bui Ngoc Bao Tran	Truong The Canh	Pham Ngoc Quyen	Nguyen Trung Dung	Tran Thu Ngan
Case Study Analysis	100%	15%	15%	15%	15%	40%
Business analysis	100%	15%	15%	40%	15%	15%
System design	100%	15%	40%	15%	15%	15%
Implementation	100%	15%	15%	15%	40%	15%
Documentation	100%	40%	15%	15%	15%	15%

7. Handover Submission:

- Source Code + Data.
- Project Introduction.
- Related Documents.